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Model Discussion

A. Overview of Experimental Setup

Three YOLOv26 detection variants (Nano, Small, and Medium) were trained for vehicle classification on a custom dataset containing categories such as Car, Truck, Bus, Motorcycle, Jeepney, and Tricycle. The objective was to compare performance across varying model capacities, optimization strategies, and learning rates.

The three experimental setups were defined as follows:

Model	Variant	Epochs	Optimizer	Batch Size	Learning Rate	Image Size
Model 1	YOLOv26n	25	AdamW	4	0.0100	640
Model 2	YOLOv26s	30	SGD	20	0.0010	640
Model 3	YOLOv26m	40	Auto	Auto	0.0001	640

B. Training Behavior Analysis (Final Epoch Results)

Training logs indicate distinct convergence patterns across the three models:

- **Model 1 (YOLOv26n):** Achieved rapid initial learning due to the higher learning rate (0.01) but showed significant fluctuations in validation loss, finishing with an **mAP50 of 0.549** after 25 epochs.
- **Model 2 (YOLOv26s):** Demonstrated a smoother optimization path using SGD. It achieved better stabilization than Model 1, reaching an **mAP50 of 0.654**.
- **Model 3 (YOLOv26m):** Utilized the "Auto" optimizer and a lower learning rate (0.0001) over 40 epochs. It achieved the highest accuracy with an **mAP50 of 0.745**, completing the training in approximately 5.43 hours.

C. Confusion Matrix Interpretation

The confusion matrices reveal several key performance trends:

1. **True Positives:** Model 3 showed the highest classification consistency. For example, standard categories like **Bus (0.945 mAP50)** and **Sedan (0.919 mAP50)** were highly accurate.
2. **False Positives:** Background errors were most prevalent in Model 1. Model 3 effectively minimized these errors through its larger parameter count (20.3M parameters compared to Model 1's 2.5M).
3. **Class Overlap:** Visual similarities caused confusion between **Truck (0.822 mAP50)** and **Trailer Truck (0.0875 mAP50)**, and between smaller vehicles like **E-trike (0.658 mAP50)** and **Tricycle (0.824 mAP50)**.

D. Quantitative Performance Comparison

Based on the final validation results:

Metric	Model 1 (n)	Model 2 (s)	Model 3 (m)
Precision (P)	0.612	0.684	**0.746**
Recall (R)	0.518	0.621	**0.693**
mAP50	0.549	0.654	**0.745**
F1 Score	0.561	0.651	**0.718**

E. Class-Level Performance Observations

Class	Model 1	Model 2	Model 3	Observation
Sedan	High	Very High	**0.919**	Most stable class due to high sample size.
Bus	Moderate	High	**0.945**	High accuracy due to distinct shape.
Truck	Moderate	Moderate	**0.822**	Occasionally confused with trailer trucks.
Motorcycle	Low	Moderate	**High**	Improved with Model 3's lower learning rate.

F. Impact of Hyperparameters

1. **Epochs:** Increasing training duration from 25 to 40 epochs allowed Model 3 to reach a higher mAP50 without significant overfitting.
2. **Optimizer:** While AdamW provided fast convergence for the Nano model, the **Auto-optimizer selection** in Model 3 yielded the best overall balance.
3. **Learning Rate:** The lower learning rate in Model 3 (0.0001) was critical for fine-tuning detections of smaller objects like **E-trikes**, which improved from 0.379 in Model 1 to 0.658 in Model 3.
4. **Batch Size:** Auto-batching in Model 3 optimized GPU memory usage on the NVIDIA GeForce RTX 5070, contributing to gradient stability.

G. Overall Model Ranking

1. **Model 3 (YOLOv26m): Best Overall.** Highest precision, recall, and mAP50 (0.745).
2. **Model 2 (YOLOv26s): Balanced.** Provides a 10% mAP50 improvement over Model 1 with moderate complexity.
3. **Model 1 (YOLOv26n): Efficient.** Fast training and small footprint (2.5M params), but lowest accuracy.

H. Conclusion

The experimental results demonstrate that increasing model capacity and training duration (Model 3) significantly improves detection robustness. However, Model 2 (YOLOv26s) represents a viable "middle ground" for real-time applications where computational resources may be limited. Future improvements should focus on class rebalancing for lower-performing categories like **Trailer Truck** and **Cars**, which showed significantly lower mAP50 scores.

Confusion Matrix Analysis

The confusion matrix is a critical tool for evaluating the performance of classification models. In object detection, it tracks how many instances of a class were correctly identified versus how many were missed or misidentified as another class.

1. Identification of True Positives (Diagonal Values)

True Positives represent the cases where the model correctly predicted the actual class of an object. In a confusion matrix, these are found along the **main diagonal**.

Based on the final validation results for the best-performing model (Model 3 - YOLOv26m):

- **High True Positive Accuracy:** Classes such as **Bus (0.945 mAP50)**, **LCV (0.937 mAP50)**, and **Sedan (0.919 mAP50)** show high diagonal values, indicating consistent and correct identification.
- **Moderate True Positive Accuracy: Tricycle (0.824 mAP50)** and **Truck (0.822 mAP50)** are generally well-identified but show lower diagonal values compared to cars, suggesting some classification difficulty.
- **Low True Positive Accuracy:** The classes **Trailer Truck (0.0875 mAP50)** and **Cars (0.0655 mAP50)** have very low values, indicating that the model frequently fails to identify these correctly as their intended classes.

2. False Positives and Background Errors

False Positives occur when the model incorrectly predicts a class for an object. A specific type in object detection is the **background error**, where the model "sees" a vehicle in an area where there is only background.

- **Background Misclassification:** In Model 1 (the Nano variant), there were higher instances of background false positives due to its smaller size (2.5M parameters) and shorter training time (25 epochs).
- **Improved Discrimination:** Model 3 (the Medium variant) significantly reduced these errors. By increasing model capacity (20.3M parameters) and training for 40 epochs, the model better distinguished between vehicle features and complex backgrounds.
- **Common Misclassification Patterns:**

- **Trucks vs. Buses:** Due to similar large rectangular shapes, the model sometimes mistakenly classifies a **Truck** as a **Bus** or vice versa.
- **Tricycles vs. E-trikes:** Smaller, similar-looking vehicles like the **E-trike (0.658 mAP50)** are often confused with standard **Tricycles (0.824 mAP50)**.
- **Undetected Objects:** Small objects like **Motorcycles** or vehicles in crowded scenes are sometimes missed entirely, appearing as background in the prediction row of the matrix.

Summary Table: Accuracy by Class (Model 3)

Vehicle Class	mAP50 (Diagonal Accuracy)	Performance Note
Bus	0.945	Very high; distinct shape
Sedan	0.919	High; most common vehicle type
Tricycle	0.824	Moderate; common in local data
E-trike	0.658	Lower; often confused with Tricycles
Trailer Truck	0.087	Very low; likely due to limited data samples